A High-Performance In-Memory Multi-Bit Adder Based on TST-MRAM

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***Abstract*—In traditional von-Neumann architecture, the memory and the arithmetic logic unit (ALU) are separated. The extra overhead caused by data transfer limits the performance of ALUs in data-intensive scenarios. Compute-in-memory (CiM) architecture based on emerging non-volatile memories (NVMs) has been proven to be effective in addressing the "memory wall" issue. However, current NV-CiM schemes primarily focus on the Boolean logic paradigm, limiting the parallelism of multi-bit computations. In this paper, we propose a high-performance in-memory multi-bit adder based on the toggle spin torque MRAM crossbar array. We use the time-based sensing amplifier to implement majority logic in the crossbar array. Based on majority logic gate, we design a multi-bit parallel-prefix adder. Compared to the multi-bit adders implemented based on Boolean logic gate, our work reduces the number of memory read/write operations and improves the computational parallelism. Moreover, the proposed multi-bit addition scheme exhibits $O(\log_2(n))$ latency and requires 6n cells for n-bit adder.**

Keywords*—Compute-in-memory, multi-bit adder, TST-MRAM, majority logic

I. Introduction

In data-intensive application like artificial intelligence (AI) and the internet of things (IoT), traditional von-Neumann architecture faces the "memory wall" issue. Limited bandwidth and frequent data transfers lead to significant power consumption and latency, limiting speed and energy efficiency of the computing systems. Compute-in-memory (CiM) architecture has been proven to break through the bottlenecks of energy efficiency and access speed [1], [2]. By shortening the distance between the arithmetic logic unit (ALU) and the memory, CiM architecture effectively increases bandwidth and reduces power consumption caused by data transmission. CiM designs based on the novel non-volatile memory (NVM) devices (e.g., ReRAM, MRAM) have garnered significant attention in both academia and industry [3], [4]. NVM-based CiM combines the advantage of near-zero static power consumption from NVM to enable high-performance computing. Toggle spin torque MRAM (TST-MRAM), one of the most promising NVMs, uses the tunneling magnetoresistance effect of the magnetic tunnel junction (MTJ) to store data. TST-MRAM provides higher write speed, lower write power consumption, and higher endurance than the previous generation of STT-MRAM [5].

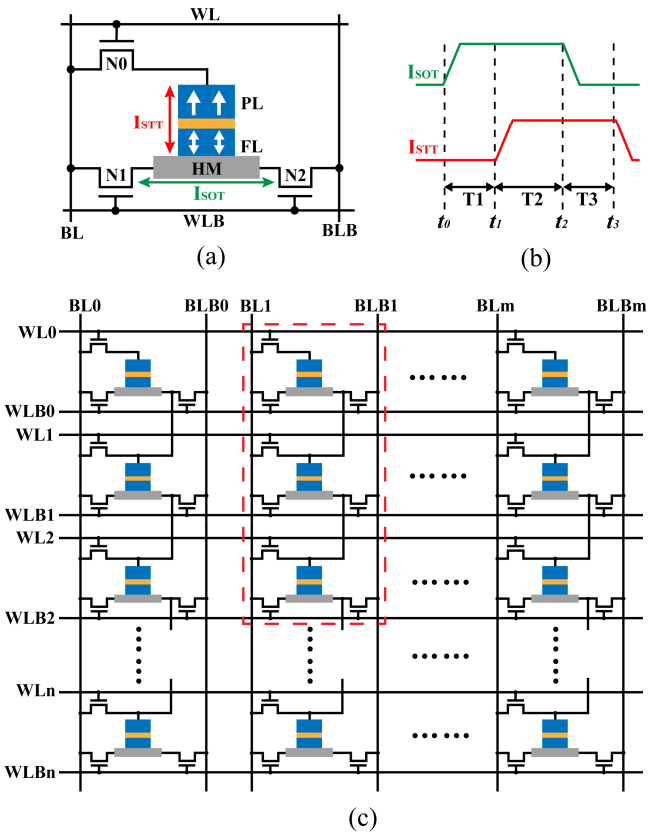


Fig. 1. (a) Structure of 3T1M TST-MRAM bit-cell, whose write operation requires both SOT current (I_{SOT}) and STT current (I_{STT}). (b) Time diagram of I_{STT} and I_{SOT} during the write operation. (c) Structure of TST-MRAM crossbar array.

The TST-MRAM crossbar array can directly perform majority logic, leveraging the resistance changes in memory cells. The CiM paradigm based on majority logic reduces the logic depth and simplifies the computation process. In this paper, we propose a high-performance in-memory multi-bit adder based on the TST-MRAM crossbar array. The time-domain sensing scheme improves the reliability of the majority

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logic operations in the TST-MRAM crossbar array. Meanwhile, when computing multi-bit addition, the proposed work effectively reduces the number of read/write operations and improves the parallelism of computation compared to adders based on the Boolean logic paradigm.

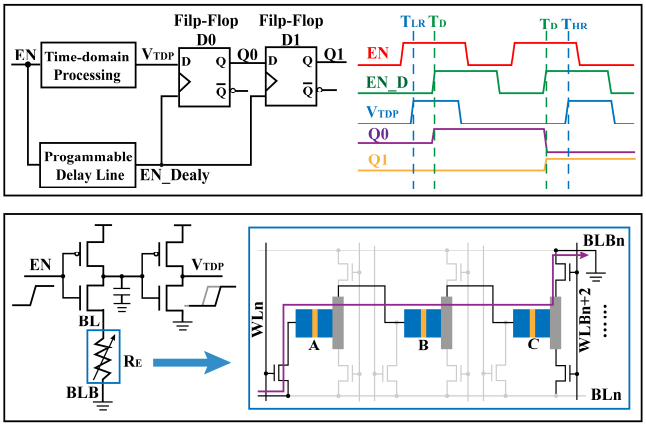
II. Background

Spin transfer torque MRAM (STT-MRAM) has been widely investigated due to its advantages of high density, high interference immunity and high speed [6]. However, the high current density during write operations in STT-MRAM accelerates the aging of the barrier layer in the MTJ, reducing device endurance [7]. The new generation spin orbit torque MRAM (SOT-MRAM) adopts a three-port read/write separated structure. By isolating the read/write paths, the problem of MTJ's aging is effectively avoided. Nevertheless, for SOT-MTJs with perpendicular magnetic anisotropy (PMA), toggle spin torque MRAM (TST-MRAM) has a mature field-free writing scheme. Fig. 1(a) shows the schematic of a standard 3T1M bit-cell of TST-MRAM, including two bit lines (BL and BLB), two word lines (WL and WLB), three NMOS transistors ($N_{0\text{-}2}$) and a triple-ported MTJ. The write operation of TST-MRAM requires the collaboration of the horizontally oriented SOT current and the vertically oriented STT current. The write timing diagram is shown in Fig. 1(b), where the SOT current firstly flows through the heavy metal layer. Subsequently the STT current flows through the MTJ nanopillar. The flow direction of the STT current determines the MTJ state. STT current flows from the free layer (FL) to the pinned layer (PL) to write the data '1', at which time the MTJ shows low resistance (R_P). The MTJ shows high resistance (R_{AP}) when it flows from the PL to the FL to write data '0' [8]-[10]. Although the storage density of TST-MRAM is lower than that of STT-MRAM, existing studies show that TST-MRAM can provide lower switching delay (~350 ps) and smaller energy dissipation (~300 fJ) when compared with STT-MRAM [11], [12].

Fig. 1(c) shows a TST-MRAM-based crossbar array. 3T1M bit-cells are connected to form columns. During the write operation, the corresponding word lines (WL_n, WLB_n) of the cell are activated. Then, the appropriate voltage is applied on the BL and BLB to write '0' or '1'. During the read operation, WL_n and WLB_{n+2} are activated. Their selected three MTJs form a series connection. This means that the resistance between BL and BLB can be seen as the sum of the resistances of the selected three bit-cells.

III. Majority Logic Gate Based on TST-MRAM Crossbar Array

Majority logic is a computing paradigm distinct from traditional Boolean logic. It has advantages over Boolean logic in area such as fault-tolerant computing and parallel processing [13]-[15]. In this paper, majority logic operations are implemented in a TST-MRAM crossbar array by using a time-based sensing amplifier (TBSA). TBSA enhances voltage and temperature robustness by reducing the reliance on current and voltage signals. This interference immunity and reduces the bit error rate. Fig. 2 illustrates the implementation of majority logic



Low resistance state		High resistance state	
A B C	R_E	A B C	R_E
(1 1 1)	$3R_p + 2R_{ON}$	(0 0 1)	$R_p + 2R_{AP} + 2R_{ON}$
(0 1 1)	$2R_p + R_{AP} + 2R_{ON}$	(0 0 0)	$3R_{AP} + 2R_{ON}$

Fig. 2. Majority gate logic based on the TST-MRAM crossbar array. The resistance R_E is detected by a time-based sensing amplifier (TBSA), consisting of a time-domain processing circuit, a programmable delay unit and two D-flip-flops.

operations in a TST-MRAM crossbar array. When WL_n and WLB_{n+2} are activated, three MTJs between them are connected in series. The resistance R_E of the path between BL and BLB can be expressed as: $R_E = R_A + R_B + R_C + 2R_{ON}$. When the CMOS transistor is turned on, its resistance R_{ON} is much smaller than that of the MTJ. Therefore, R_E can be considered as the total resistance of the three series-connected MTJs. When ABC = "111" or ABC = "011", at least two MTJs in the path (from BL to BLB) are in parallel state, resulting in a low resistance ($\leqslant 2R_P + R_{AP}$) in the path. Conversely, when ABC = "001" or ABC = "000", at least two MTJs are in anti-parallel state, resulting in a high resistance ($\geqslant R_P + 2R_{AP}$). The time-domain processing circuit in the TBSA translates the resistance signal into a time-domain signal. The time-domain processing circuit consists of two stages of inverters, with the source of the NMOS transistor in the first stage connected to BL. During the read operation, the rising edge pulse passes through the time-domain processing circuit, generating a delayed signal. According to the RC circuit model, the delay time is positively correlated with R_E. D flip-flop detects the delay time and outputs majority logic results.

Considering the process variations of the CMOS devices and MTJ, the results of Monte Carlo (MC) simulations of the TBSA are shown in Fig. 3. As observed in Fig. 3(a), process variations affect the delay time generated by the time-domain processing circuit. The worst-case time window is approximately 1.5 ns when ABC = "011" and ABC = "001". Fig. 3(b) demonstrates the TBSA output results, showing a clear distinction between R^E_{011} and R^E_{001}, which can be reliably detected by the TBSA.

IV. Proposed High Performance Multi-Bit Adder Based on Majority Logic Gate

Implementing multi-bit addition by using majority logic gates offers the advantage of high speed. In a single-bit full

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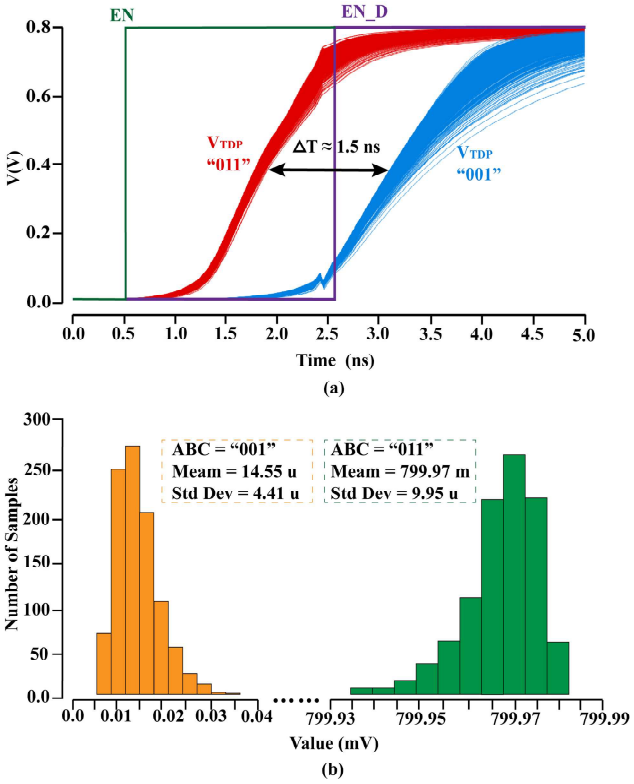


Fig. 3. (a) Sample waveforms for MC simulation of TBSA. The time-domain processing circuit circuit has a worst-time difference of 1.5 ns between the output waveforms at ABC = "001", "011". (b) MC simulation diagram of the proposed majority logic gate result at ABC = "001", "011".

adder, the output carry C_i and the output sum S_i can be represented by using majority gates, shown in Eq. 1 and Eq. 2. It is clear that S_i only depends on input data and carry signals. Therefore, the speed for implementing an N-bit adder is primarily determined by the carry generation block.

$$C_{i+1} = \mathrm{Maj}\ (a_i, b_i, C_i) \tag{1}$$

$$S_i = \mathrm{Maj}\ (\overline{C_{i+1}}, \mathrm{Maj}\ (a_i, b_i, \overline{C_{i+1}}), C_i) \tag{2}$$

Since majority gate logic is applicable to many emerging nanotechnologies, especially resistive memory devices such as ReRAM and MRAM. Significant efforts have been directed toward designing various parallel-prefix adders using majority logic. However, considering the larger area of the 3T1M cell in TST-MRAM, this work proposes a multi-bit adder scheme optimized for the TST-MRAM crossbar array. The scheme effectively multiplexes column data during the carry generation phase， saving array area while achieving high performance.

As shown in Fig. 4(a), the block diagram of an N-bit parallel-prefix adder consists of a carry generation block and a sum generation block. After the carry generation block produces its output, the sum generation block performs the summation operation in parallel. Fig. 4(b) illustrates the improved carry generation block of an 8-bit parallel-prefix adder. Unlike the standard Ladner Fischer adder, which uses a carry generation block with four layers [13], the improved version incorporates five layers. Each layer contains four majority logic gates. The

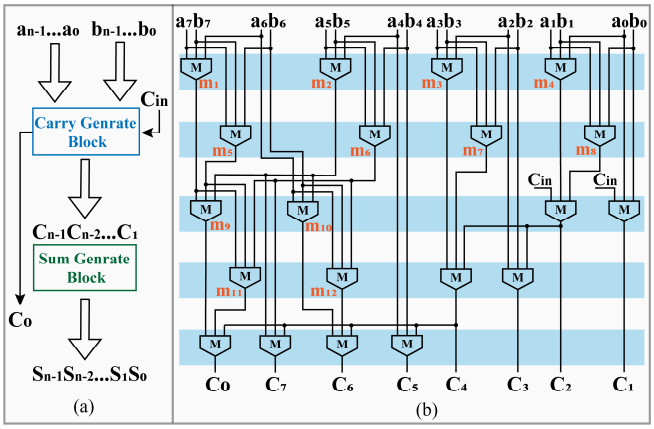


Fig. 4. (a) Block diagram of N-bit parallel-prefix adder. (b) The improved carry generate block of an 8-bit parallel-prefix (Ladner Fischer) adder can be represented by a 5-level majority gate.

gates within the same layer operate in parallel. It should be noted that the more majority gates in each layer the more cells are required, amplifying the disadvantage of TST-MRAM's large cell area. We strategically increased the number of layers in the carry generation block and evenly distributed the majority logic gates across each layer. This approach enhances area efficiency in TST-MRAM.

Fig. 5 presents the mapping of the proposed 8-bit parallel-prefix adder structure to the TST-MRAM crossbar array. Initially, all cells are set to '0', except for the ones that store data. The row decoder restricts the selection to only three consecutive rows for majority logic operations. Furthermore, because the write state of TST-MRAM is dictated by the direction of the I_{STT}, write operations cannot be performed simultaneously on multiple rows in the same column. This limitation results in more write steps compared to read steps during computation. The steps for calculating the 8-bit parallel-prefix addition are outlined below:

1) Majority at col. (2, 4, 6, 8) of row (2, 3 ,4);
2) Majority at col. (2, 4, 6, 8) of row (1, 2 ,3);
3) Write (m1, m2, m3, m4) at col. (1, 3, 5, 7) of row4;
4) Write (m5, m6, m7, m8) at col. (1, 3, 5, 7) of row5;
5) Write (m2, m2, Cin, Cin) at col. (1, 2, 7, 8) of row6;
6) Majority at col. (1, 2, 7, 8) of row (4, 5 ,6);
7) Write (m6, m6, C2, C2) at col. (1, 2, 5, 6) of row6;
8) Majority at col. (1, 2, 5, 6) of row (4, 5 ,6);
9) Write (m9, m10) at col. (1, 2) of row4;
10) Write (m11, m12) at col. (1, 2) of row5;
11) Write (C4, C4, C4, C4) at col. (1, 2, 3, 4) of row6;
12) Majority at col. (1, 2, 3, 4) of row (4, 5 ,6);
13) Write (/Co, /C7, /C6, /C5, /C4, /C3, /C2, /C1) at col. (2, 3, 4, 5, 6, 7, 8, 9) of row4;
14) Write (C7, C6, C5, C4, C3, C2, C1, Cin) at col. (2, 3, 4, 5, 6, 7, 8, 9) of row5;
15) Majority at col. (2, 3, 4, 5, 6, 7, 8, 9) of row (2, 3 ,4);

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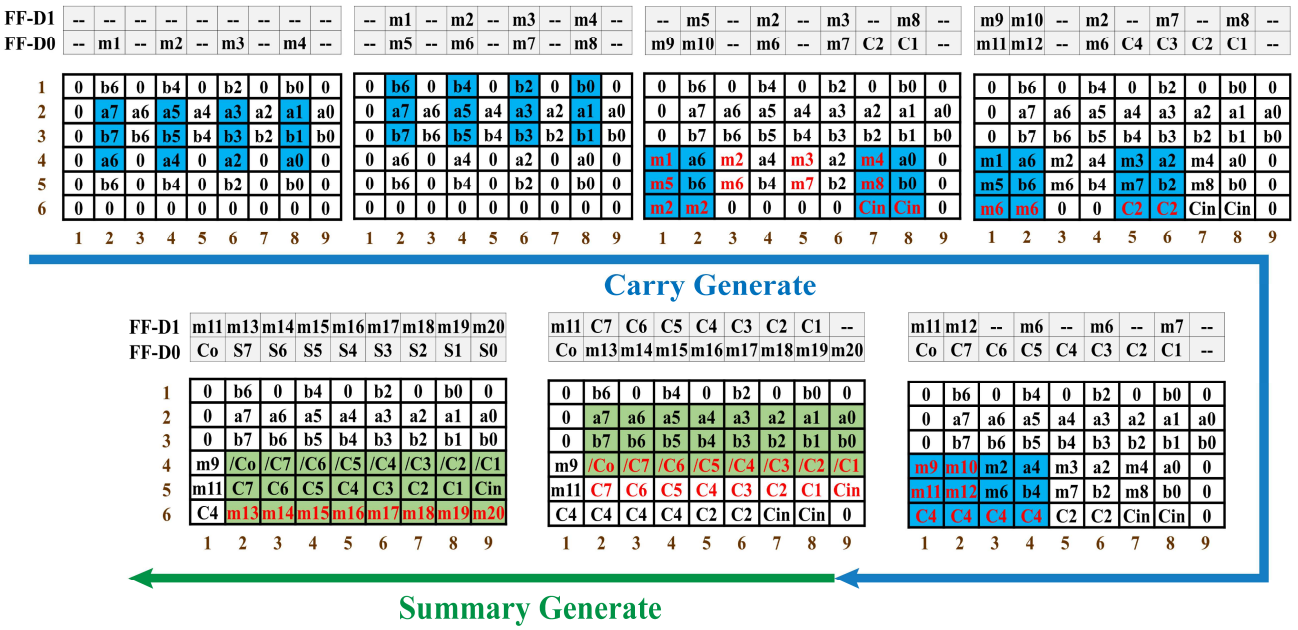


Fig. 5. Mapping of the TST-MRAM crossbar array to the 7 stages of the 8-bit parallel-prefix adder. Level 1 to level 5 correspond to carry operation, level 6 and level 7 correspond to sum operation. Each stage performs write operations first (highlighted in red) and majority logic operations second (shaded in blue or green). Flip-flops D0 and D1 (shaded in grey) can store the results of two consecutive majority logic calculations. Each level of majority logic corresponds to TST-MRAM crossbar array of size 6×9.

16) Write (m13, m14, m15, m16, m17, m18, m19, m20) at col. (2, 3, 4, 5, 6, 7, 8, 9) of row6;

17) Majority at col. (2, 3, 4, 5, 6, 7, 8, 9) of row (4, 5 ,6).

The proposed 8-bit parallel-prefix adder requires 17 steps, consisting of 36 majority logic operations and 48 write operations. Additionally, the adder multiplexes column data, resulting in a smaller array area occupancy of 54 cells.

Based on the previous analysis of the 8-bit parallel-prefix adder, we derive the number of steps and cells required to execute an N-bit adder. An N-bit adder requires a total of $6\log_2(n) - 1$ steps. The carry generate block requires $6\times(\log_2(n) - 1)$ steps, while the sum generate block requires 5 steps. The N-bit adder occupies a total of 6(n+1) cells in the TST-MRAM crossbar array. Tab. I presents a comparison with other designs. In terms of speed, due to the fast write speed of TST-MRAM, the latency of our proposed adder is 5 ns. Furthermore, the high parallelism of our parallel-prefix adder results in fewer computational steps for high-bit additions compared to designs in [16], [17] and [18]. In terms of cell overhead, our proposed design requires only two-thirds, one-third, and one-eighth of the cell area compared to [16], [17] and [19], respectively.

V. Conclusion

In this paper, we proposed a high-performance multi-bit adder based on a TST-MRAM crossbar array. We implemented a highly reliable majority logic operation in the TST-MRAM crossbar array using a time-based sensing amplifier (TBSA). Additionally, a multi-bit parallel-prefix adder scheme was proposed. We appropriately multiplexed the column data to strike a balance between the area efficiency of the adder and its parallelism. Implementing this scheme in the TST-MRAM crossbar array requires $6\log_2(n) - 1$ steps and occupies an array area of 6(n+1) cells.

TABLE I. Comparison With Other Works

Name	Steps			Cell Num.	
	Latency	N-bit	8-bit	N-bit	8-bit
[16]	50 ns	2N + 2	900 ns	9N + 43	115
[17]	5 ns	3N + 3	135 ns	18N	144
[18]	6 ns	4N + 1	198 ns	2N + 9	25
[19]	100 ns	$4\log_2(n) + 6$	1800 ns	48(N + 2)	480
Our work	5 ns	$6\log_2(n) - 1$	85 ns	6(N + 1)	54

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